

Magnetic Position Sensor Report

This report documents the performance evaluation of a precision rotary axis with different encoders, namely [MLX90382](#) (**Hall effect**, 16bit res) and [TLE5012B](#) (**GMR based**, 15bit res)

The algorithm used is same for both sensors, just a Simple Moving Average with 1000 samples. Main goal is to check accuracy, error range(peak to peak error) and repeatability.

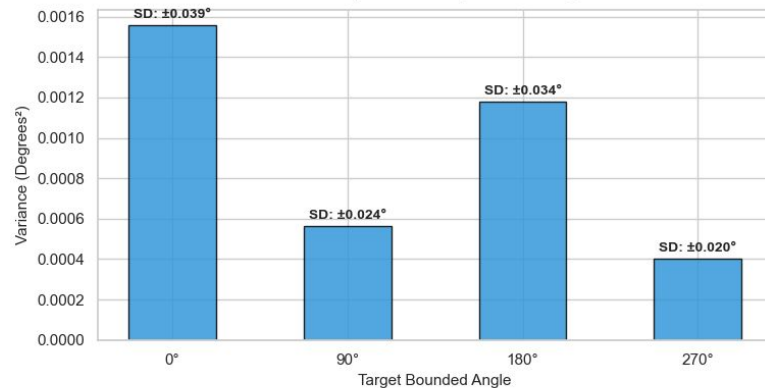
Stepper moves from 0-90-180-270-0, for 500 cycles, values are logged and plotted for both encoders,

Same test is done but with 30 degree increments too, 0-30-60....330-0.

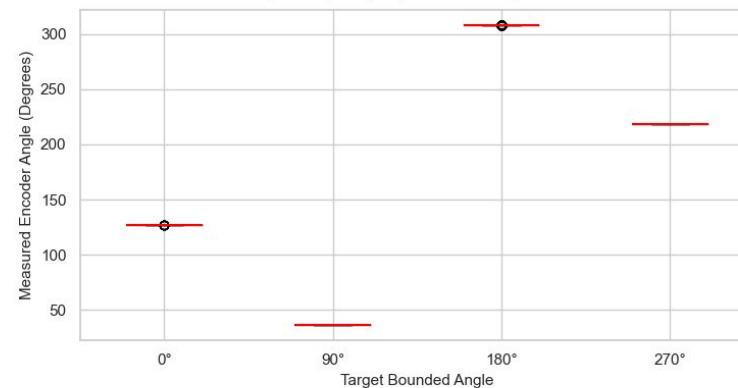
Variance at each angle, boxplot and Probability Density Function is plotted for all the tests.

MLX90382 Linear Axis Repeatability Report (500 Complete Cycles)

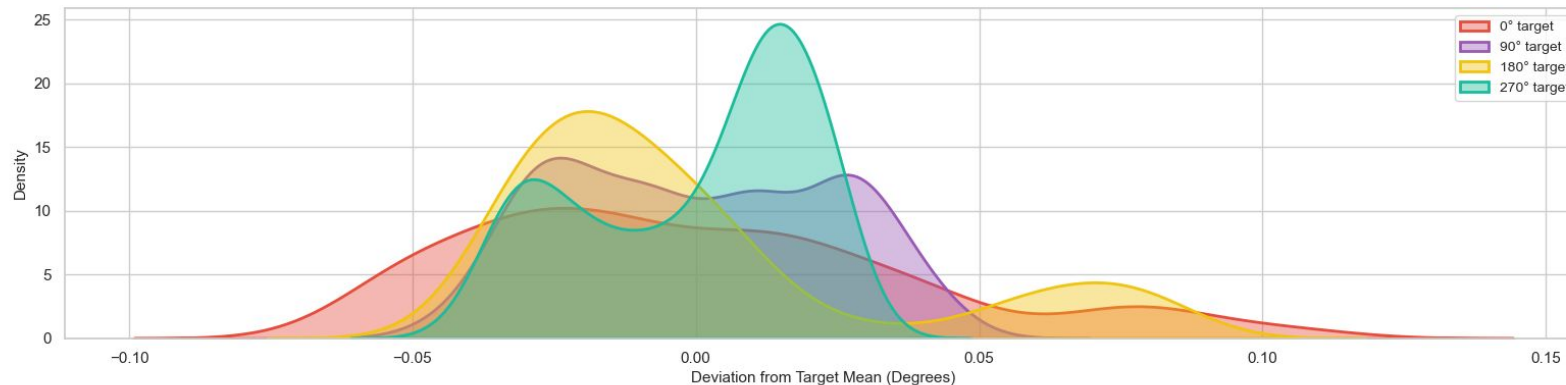
Variance (Noise Floor) at Each Angle



Full Spread (Boxplot) at Each Target Position



Overlay of Measurement Noise Distributions (Mean Centered to 0.0°)



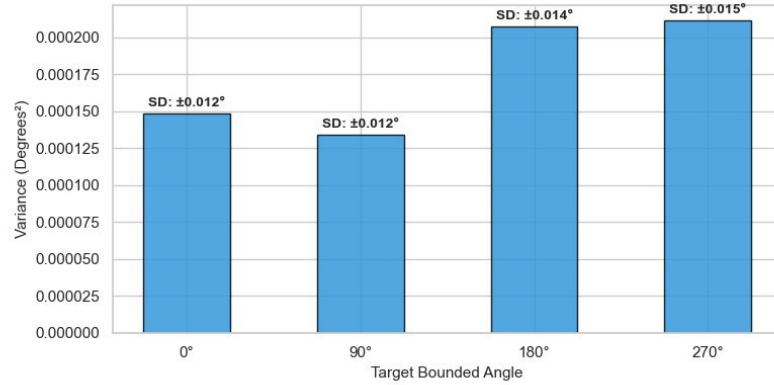
Target (mm/deg)	Min Value	Max Value	Delta (Max-Min)	Variance
0° (Supposed)	127.189°	127.364°	0.1752°	0.001555
90° (Supposed)	36.456°	36.554°	0.0975°	0.000565
180° (Supposed)	308.156°	308.291°	0.1343°	0.001177
270° (Supposed)	218.155°	218.229°	0.0748°	0.000401

1. MLX90382 (Hall) 4-Step Sweep

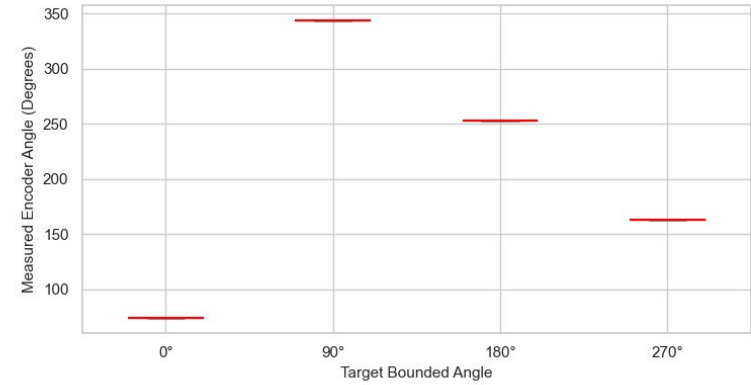
- **Results:** Repeatability error of **0.075° to 0.175°** | Variance of **0.00040 to 0.00155**.
- **Inference:** Mechanical backlash is present. Unequal error at different angles. Mechanical mounting can be better, decent resolution.

TLE5012B Linear Axis Repeatability Report (500 Complete Cycles)

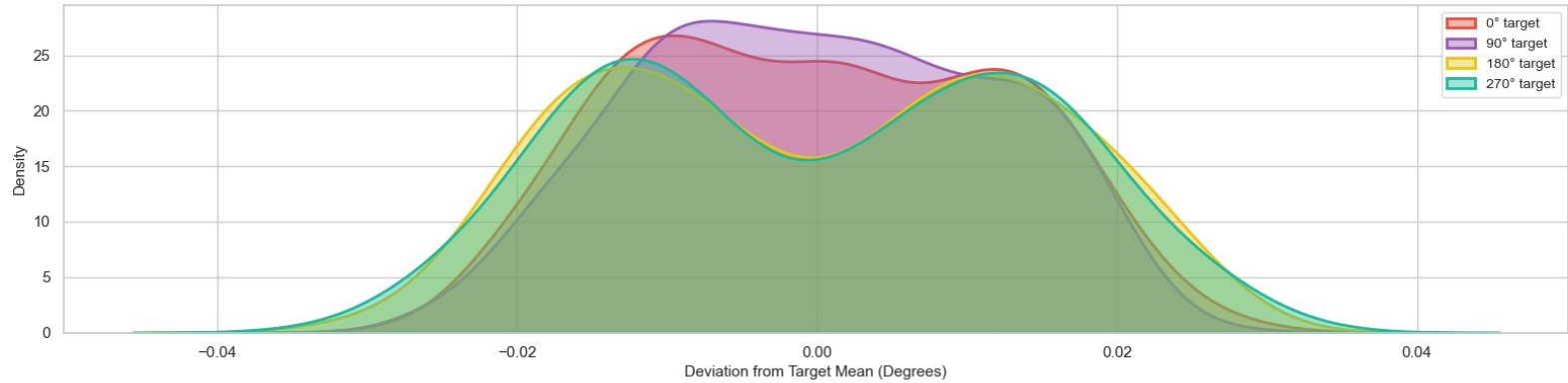
Variance (Noise Floor) at Each Angle



Full Spread (Boxplot) at Each Target Position



Overlay of Measurement Noise Distributions (Mean Centered to 0.0°)



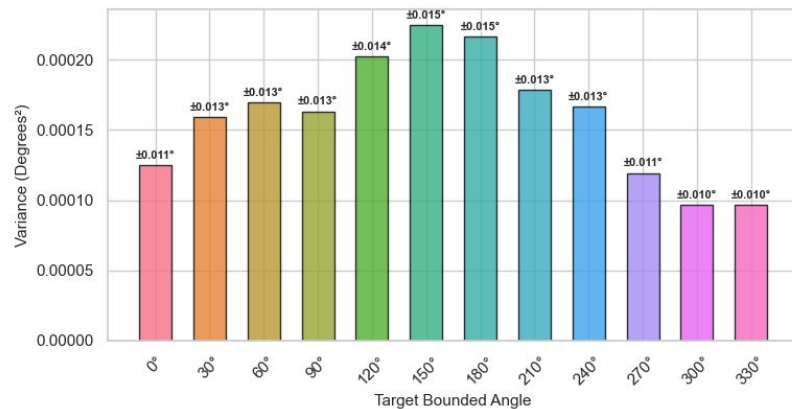
#	Target (mm/deg)	Min Value	Max Value	Delta (Max-Min)	Variance
#	0° (Supposed)	73.851°	73.909°	0.0580°	0.000148
#	90° (Supposed)	344.148°	344.202°	0.0540°	0.000134
#	180° (Supposed)	253.128°	253.189°	0.0610°	0.000207
#	270° (Supposed)	162.777°	162.843°	0.0660°	0.000211
#					

2. TLE5012B (GMR) 4-Step Sweep

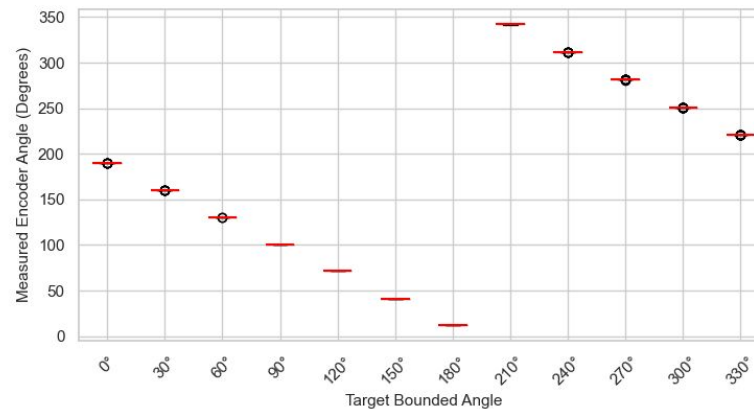
- **Results:** Repeatability error of **0.054° to 0.066°** | Variance of **0.00013 to 0.00021**.
- **Inference:** Incredibly clean and stable. Because GMR is immune to slight mounting offsets, this proves the actual physical backlash is tiny, only **~0.06°**.

TLE5012 Rotary Axis 30 Degree Repeatability Report (500 Complete 12-Step Cycles)

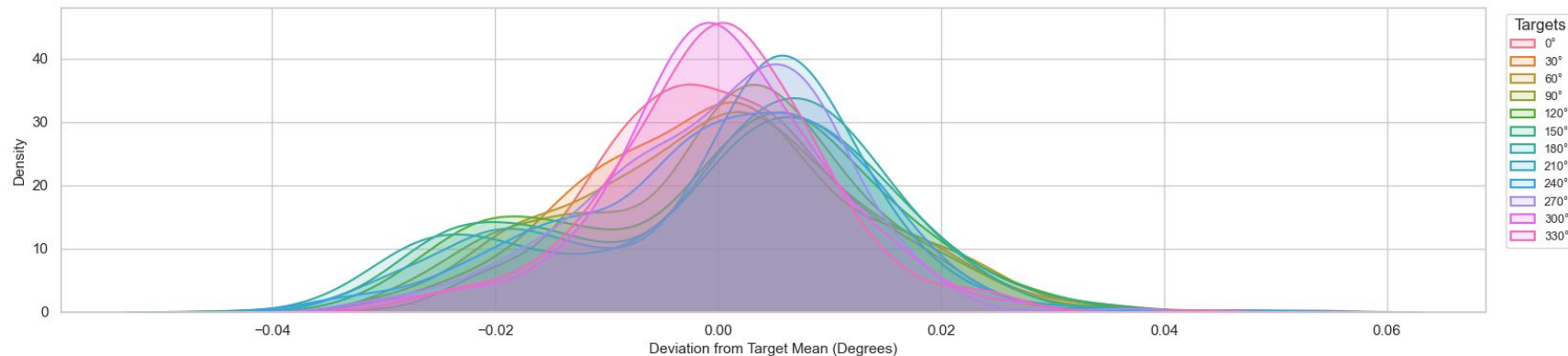
Variance (Noise Floor) at Each Angle



Full Spread (Boxplot) at Each Target Position



Overlay of Measurement Noise Distributions (Mean Centered to 0.0°)



TLE5012 Rotary Axis 30 Degree Repeatability Report (500 Complete 12-Step Cycles)

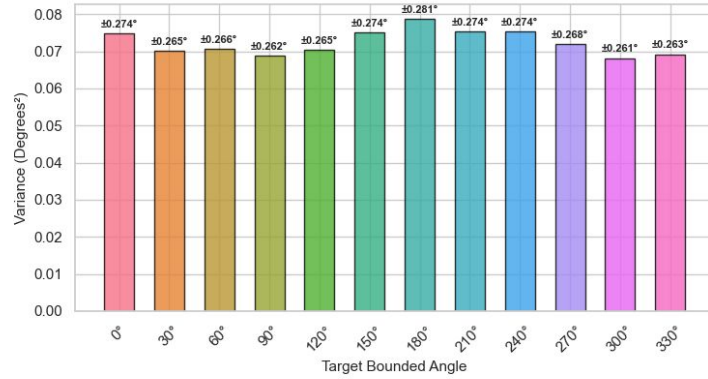
Target (deg)	Min Value	Max Value	Delta (Max-Min)	Variance
0° (Supposed)	190.007°	190.101°	0.0940°	0.000125
30° (Supposed)	160.335°	160.421°	0.0860°	0.000159
60° (Supposed)	130.598°	130.671°	0.0730°	0.000169
90° (Supposed)	100.858°	100.923°	0.0650°	0.000163
120° (Supposed)	71.272°	71.337°	0.0650°	0.000202
150° (Supposed)	41.334°	41.409°	0.0750°	0.000225
180° (Supposed)	11.617°	11.687°	0.0700°	0.000216
210° (Supposed)	341.915°	341.985°	0.0700°	0.000179
240° (Supposed)	311.494°	311.581°	0.0870°	0.000167
270° (Supposed)	281.130°	281.201°	0.0710°	0.000119
300° (Supposed)	251.102°	251.161°	0.0590°	0.000097
330° (Supposed)	220.451°	220.524°	0.0730°	0.000097

3. TLE5012B (GMR) — 12-Step Sweep

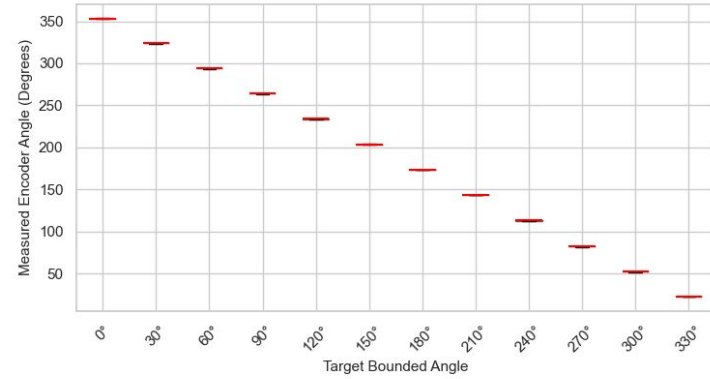
- **Results:** Repeatability error of **0.059° to 0.094°** | Variance of **0.00009 to 0.00022**.
- **Inference:** Excellent consistency.

MLX Rotary Axis 30 Degree Repeatability Report (500 Complete 12-Step Cycles)

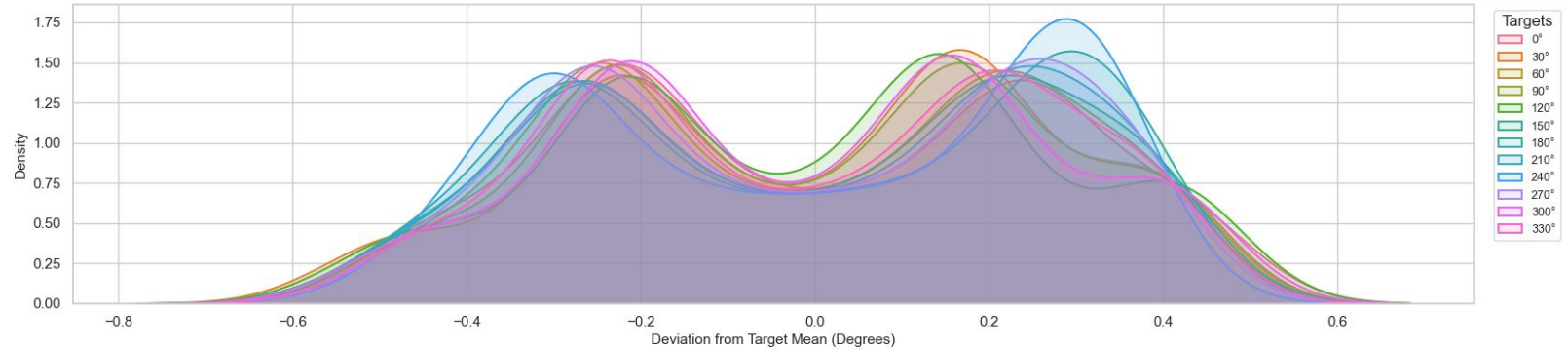
Variance (Noise Floor) at Each Angle



Full Spread (Boxplot) at Each Target Position



Overlay of Measurement Noise Distributions (Mean Centered to 0.0°)



MLX Rotary Axis 30 Degree Repeatability Report (500 Complete 12-Step Cycles)

Target (deg)	Min Value	Max Value	Delta (Max-Min)	Variance
0° (Supposed)	352.558°	353.547°	0.9885°	0.074892
30° (Supposed)	323.283°	324.254°	0.9704°	0.070156
60° (Supposed)	293.639°	294.544°	0.9041°	0.070678
90° (Supposed)	263.495°	264.434°	0.9385°	0.068752
120° (Supposed)	233.357°	234.343°	0.9856°	0.070383
150° (Supposed)	203.171°	204.105°	0.9343°	0.075052
180° (Supposed)	173.018°	173.939°	0.9203°	0.078725
210° (Supposed)	142.976°	143.900°	0.9239°	0.075252
240° (Supposed)	112.544°	113.395°	0.8503°	0.075309
270° (Supposed)	81.974°	82.869°	0.8953°	0.071925
300° (Supposed)	51.886°	52.852°	0.9663°	0.068173
330° (Supposed)	22.174°	23.091°	0.9167°	0.069227

4. MLX90382 (Hall) — 12-Step Sweep

- **Results:** Repeatability error of **0.850° to 0.988°** (Nearly 1°!) | Variance of **0.068 to 0.078**.
- **Inference:** good PDF, but huge offsets on either side. Variance is higher compared to GMR